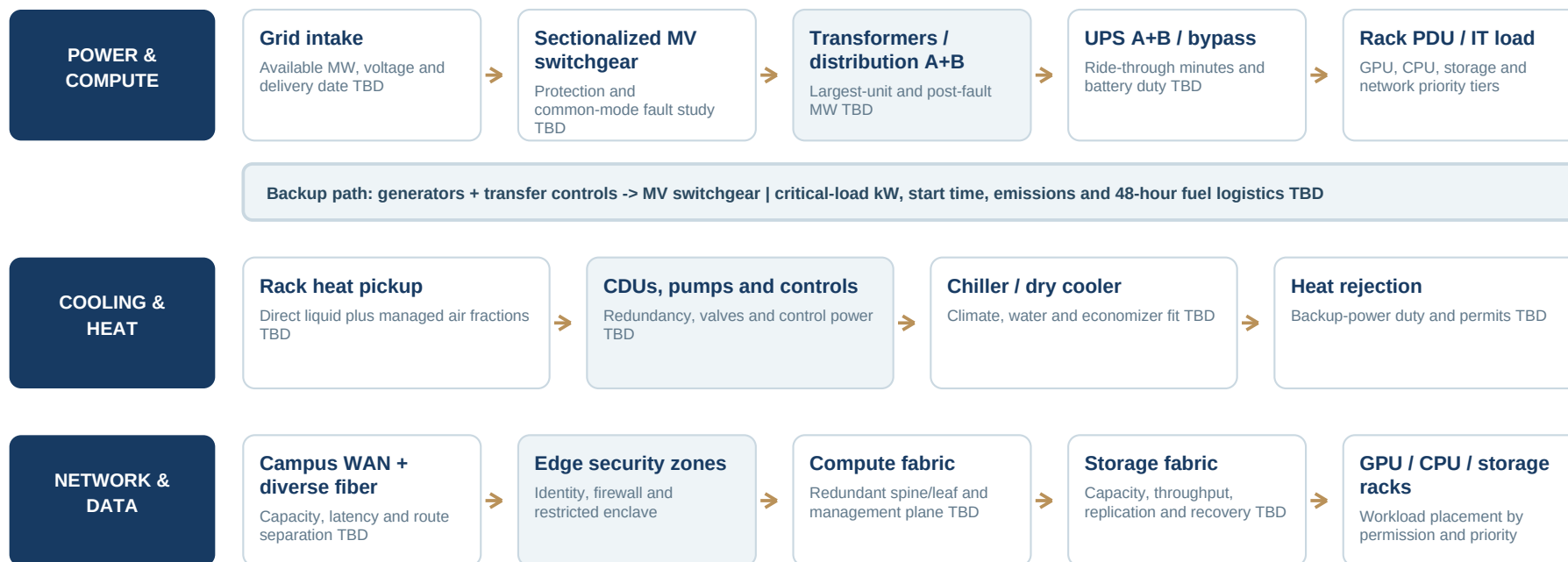


25 MW AI Data Center - Concept System Architecture

Power | cooling | network and data | largest-component fault | 48-hour outage

Planning envelope: 25 MW facility input
Assignment baseline: 20 MW IT at PUE 1.25
Preferred staged case: 12 MW -> 15 MW



FAILURE PATHS AND OPERATING RESPONSE

FAULT CASE

Largest transformer or switchgear segment fails

UPS bridges switching only. Preserve storage, security and necessary inference; checkpoint and pause batch work if the surviving path cannot carry all load. Full 25 MW after one fault is not yet proven.

FAULT CASE

Grid unavailable for 48 hours

Generators, fuel delivery and cooling controls support the defined critical load. Stop interruptible batch work and shift only approved workloads externally. Full-load 48-hour operation is not claimed.

ENGINEERING GATE BEFORE CAPITAL APPROVAL

Obtain a peak IT/cooling load sheet; one-line electrical study; largest-unit and post-fault capacity; UPS minutes; generator critical kW and fuel plan; rack liquid/air mix; water and heat-rejection study; diverse WAN/fabric topology; protection coordination; commissioning and recovery procedures.